

Numerical Investigation of the Aerodynamic and Acoustical Properties of a Shrouded Rotor

I. V. Abalakin^a, V. A. Anikin^b, P. A. Bakhvalov^a,
V. G. Bobkov^a, and T. K. Kozubskaya^a

^a*Keldysh Institute of Applied Mathematics, Miusskaya pl. 4, Moscow, 125047 Russia*

^b*Design office "Kamov", ul. 8 Marta 8-A, Lyubertsy, Moscow oblast, 111625 Russia*

e-mail: veld13@gmail.com

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Abstract—The problem of shrouded rotor rotation is numerically solved basing on the Navier–Stokes equations in a noninertial coordinate system. The configuration considered is a model helicopter tail rotor. The calculations are performed using edge-based reconstruction (EBR) schemes on unstructured tetrahedral grids with the variables determined at gridpoints. The numerical results on the aerodynamic forces and the acoustic radiation intensity and direction in the far field are presented and analyzed.

Keywords: *shrouded rotor, viscous gas flows, aerodynamic forces, far field acoustics, numerical modeling, rotating coordinate system, EBR schemes, unstructured grids.*

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In developing modern helicopters the greatest possible amount of information on airflow around the vehicle and its possible effect on the structure is of key importance for optimizing its aerodynamic properties, such as the thrust and the lift, the fuel efficiency, the flight velocity, etc. Moreover, this information must be accessible for different flight regimes, which can be hardly realizable in the ground-based experiments, while the conduction of in-flight experiments in series is limited owing to their expensiveness. The acoustical parameters of a vehicle regimented by ever- more-demanding norms concerning the permissible environmental noise adopted by ICAO Assemblies for both aircraft and helicopters also become increasingly more important. In this situation the mathematical modeling on high-performance computational complexes turns out to be very helpful in obtaining the necessary data; at present it is increasingly more actively promoted into engineering investigations in different fields of industry.

The problems to be solved in mathematically describing and calculating flows around flight vehicles are very complicated. Particularly serious difficulties arise in numerically modeling unsteady parameters of turbulent flows. As applied to the helicopter engineering, the situation is even more difficult in view of the necessity of taking account for the rotational motion of helicopters rotors. This suggests that, in view of the necessity of taking account for many interrelated factors, the problem of the development of the computational technologies for modeling the aerodynamic and acoustical parameters of a helicopter as a whole requires serious theoretical studies. The effectiveness of this methodology and its suitability in even estimative calculations can be evaluated only by means of thorough testing and comparison with the available experimental data.

In this study, the adequacy of the numerical method of the calculation of a shrouded rotor developed by the authors is confirmed on the basis of the simulation of the aerodynamic and acoustical parameters of a rotor in a ring-shaped channel in the absence of an external flow followed by their comparison with the results of an experiment conducted on a setup consisting of four-blade rotor mounted in a contoured channel in the hovering operation regime (Fig. 1a).

There are only a few studies devoted to similar configurations. Of the available publications we can distinguish papers [1, 2], which present the theory and the methods of approximate analytical calculations

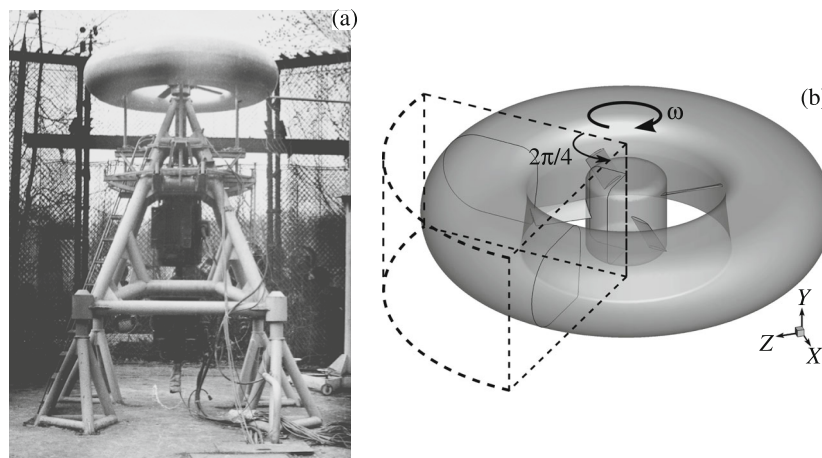


Fig. 1. General configuration of the model shrouded rotor; experimental setup (a) and computational model of the rotor with the sector under consideration containing one blade (b).

of the shrouded rotor aerodynamics, and [3], together with the bibliography therein, devoted to the numerical modeling of the aerodynamic and acoustical parameters of such systems. Original numerical methods for calculating the helicopter rotor parameters are also described in [4–6]. Thus, in [4] a numerical technique for modeling the high-speed helicopter noise is considered in a simplified formulation on a structured grid. The advantage of that study is that it considers the entire sequence of the numerical experiment up to the calculation of the tonal noise in the far field. In [5] the aerodynamic characteristics of the rotor were determined by numerically solving the system of Euler equations on arbitrary unstructured grids within the framework of the finite-volume approach. The Ffowcs Williams–Hawkings (FWH) method [6] was employed for calculating the aerodynamic characteristics of the rotor.

In this paper we present the results of a numerical investigation of the model problem of shrouded rotor under small assumptions. In particular, the channel is a body of revolution, while the rotor combines a central body (body of revolution) and the rotor blades which are of the same shape, have the same angle of pitch, and are located in the same plane (plane of rotor rotation). This formulation makes it possible to calculate only one sector of the shrouded rotor configuration imposing periodic boundary conditions on the lateral planes of the sector in the azimuthal direction. The validation is performed against the available experimental data on the aerodynamic characteristics of rotors. In this study, we also investigate the acoustic properties of the model tail rotor under consideration.

The distinctive feature of the numerical method presented in this study is the use of a cost-effective higher-accuracy method with the determination of the variables at the gridpoints of unstructured tetrahedral grids. As distinct from [5], the mathematical model describing the rotor aerodynamics is based on the system of Navier–Stokes equations for a compressible gas written in a noninertial reference frame fitted to rotor rotation. The use of unstructured grids makes considerably easier the procedure of constructing grid models adaptable to variations in the helicopter and, in particular, rotor, geometry. The high accuracy of the numerical results obtained on these grids is ensured by the application of the EBR scheme [7–9] based on the quasi-one-dimensional edge-based approximation of inviscid fluxes.

1. PROBLEM OF VISCOUS FLOW AROUND A SHROUDED FOUR-BLADE ROTOR

The geometry of the model under consideration is based on an actual experimental setup representing a shrouded four-blade tail rotor. In the channel there is a single rotor without details of its mechanization, while the channel (ring), where the rotor rotates, represents an axisymmetric body of revolution. The central body is in the shape of a cylinder with rounded end faces.

As mentioned above, due to the symmetry of this formulation in the absence of external flow the problem is solved for a sector with the angle $2\pi/N$ ($N = 4$) containing one rotor blade (Figs. 1b and 2a), the periodic

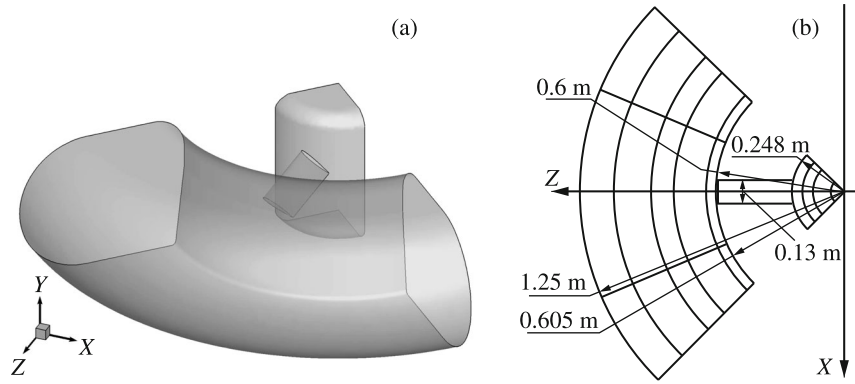


Fig. 2. Sector with one blade; geometry (a) and dimensions (b).

boundary conditions being imposed on the lateral planes of the sector in the azimuthal direction.

In accordance with the description of the experimental setup, the radius of the central body, to which the blade is fastened, is taken to be 0.248 m, the rotor radius being $R = 0.6$ m. The blade chord $b = 0.13$ m. The blade surface is based on the TsAGI SV-11 airfoil with the linear twist $\varphi_{sw}(r) = 40(0.7 - r/R)/3$, where r is the distance to the axis of rotation of the rotor. The outside channel radius is 1.25 m, while the inside channel radius in the plane of rotation of the rotor is 0.605 m, which corresponds to a 5 mm gap between the blade end and the inside surface of the channel (Fig. 2b). We consider the rotor rotation regime at a frequency of 19.44 Hz, which corresponds to the angular velocity $\omega = 122.17$ rad/s and the linear velocity of the blade end $V_{blade} = \omega R = 73.3$ m/s; the angle of pitch of the blade φ can take the values of 10, 15, 20, 30, and 40°. The Reynolds number is determined as $Re = \rho_0 V_{blade} b / \mu_0$ and takes the value $Re = 6.28 \times 10^5$ for the values $\rho_0 = 1.2041$ kg/m³ and $\mu_0 = 1.827 \times 10^{-5}$ N s/m² corresponding to the air density and dynamic viscosity at the temperature of 20°C.

2. THE PROBLEM CHARACTERISTICS UNDER STUDY

1. *Aerodynamic forces of the shrouded rotor.* In the problems of the type described above the main parameters characterizing the properties of the configuration under study are the coefficients of aerodynamic forces, namely, the thrust and torque coefficients.

We will determine the aerodynamic forces of the blade and the segment of the surrounding ring following the book [10]. For this purpose, we will consider two reference frames (Fig. 3), namely, the fixed coordinate system (x, y, z) and a coordinate system (x', y', z') fitted to the rotatable blade and obtained by rotation of the fixed coordinate system by an angle $\psi(t) + \pi/2$, where the angle $\psi(t)$ is determined by the angular velocity of blade rotation: $\psi(t) = \omega t$.

Since the calculations are made in the rotatable coordinate system, including the case of stationary solution (the dependence on the azimuthal angle ψ is absent), the axes of the fixed and rotatable coordinate systems can be identified.

Let $p(x, y, z)$ be the pressure distribution over the blade surface S and n_x, n_y , and n_z be the components of the unit normal outward with respect to the surface S . Then the N -bladed rotor thrust T_{rot} is composed of the forces T_y^{blade} normal to the plane of rotation and acting on single blades

$$T^{rot} = N T_y^{blade} = N \int_S p n_y ds. \quad (2.1)$$

The rotor force moment is determined by the force projections on the axes of the fixed coordinate system and the radius-vector (lever) of the blade coordinates in the rotatable coordinate system $\mathbf{r} = (0, 0, z) = (0, 0, z')$

$$\mathbf{M} = N \int_S p (\mathbf{r} \times \mathbf{n}) ds.$$

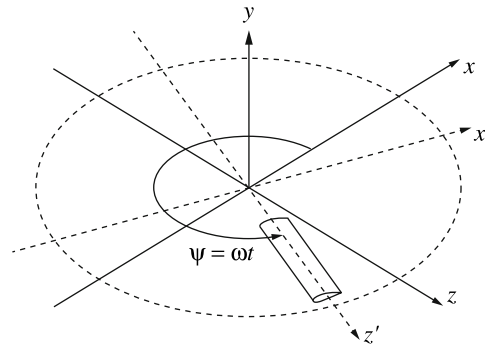


Fig. 3. Fixed and rotatable coordinate systems.

Hence there follows the expression for the aerodynamic torque of the rotor

$$M_k^{rot} = M_y = N \int_S z p n_x ds.$$

In the case of the stationary solution the averaging over the azimuthal angle is not made in the above expressions for the thrust and the torque.

Let now $p(x, y, z)$ be the pressure distribution over the surface of a ring sector with the angle of $2\pi/N$. This pressure distribution corresponds to the fixed blade position near the ring at a certain moment of time (the pressure distribution in the rotatable coordinate system). For this reason, to determine the pressure distribution in the fixed coordinate system requires averaging the pressure over the azimuthal angle ψ . In the case of steady flow the angle-average pressure is the same for any of N segments. Then, passing to the cylindrical coordinates (r, ψ, z) we can calculate the azimuthal-angle-average pressure

$$\langle p \rangle(r, z) = \int_0^{2\pi} p(r, \psi, z) d\psi.$$

The ring thrust is calculated from the formula analogous to (2.1)

$$T^{ring} = N \int_S \langle p \rangle n_y ds,$$

where S is the ring surface and n_x, n_y , and n_z are the components of the unit normal outward with respect to the ring surface. The ring torque is determined by the force projections on the axes of the fixed coordinate system and the radius-vector of the ring coordinates in the fixed coordinate system $\mathbf{r} = (x, y, z)$

$$M_k^{ring} = N \int_S \langle p \rangle (zn_x - xn_z) ds.$$

The thrust coefficient c_T and the aerodynamic torque coefficient m_k are determined by nondimensionalizing the thrust and torque values by the quantities $\rho_0 A (\omega R)^2 / 2$ and $\rho_0 R A (\omega R)^2 / 2$, respectively

$$c_T^{rot/ring} = \frac{2T^{rot/ring}}{\rho_0 A (\omega R)^2}, \quad m_k^{rot/ring} = \frac{2M_k^{rot/ring}}{\rho_0 R A (\omega R)^2},$$

where ρ_0 is the undisturbed flow density, $A = \pi R^2$ is the blade disk area, R is the blade radius, and ω is the absolute value of the angular velocity of the blade.

The values of the calculated force coefficients are compared with those obtained in the full-scale experiments.

2. *Acoustical characteristics of the shrouded rotor.* An important acoustical characteristic of a flow is the overall sound pressure level (OASPL) in the far field which is determined by the far-field pressure fluctuation spectrum $P(\mathbf{x}, f)$. The pressure fluctuation spectrum is given by the Fourier transform of the function $p'(\mathbf{x}, t)$ normalized by the quantity $p_0 = 2 \times 10^{-5}$ Pa

$$P(\mathbf{x}, f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{p(\mathbf{x}, t)}{p_0} e^{-i2\pi f t} dt.$$

We will determine the total energy of the spectrum as the integral of the spectral power $S(\mathbf{x}, f) = P^2(\mathbf{x}, f)$

$$E(\mathbf{x}) = \int S(\mathbf{x}, f) df,$$

where the integration is performed over all resolvable frequencies. Then the OASPL expressed in decibels is determined from the formula

$$\text{OASPL} = 10 \log_{10} E(\mathbf{x}).$$

3. MATHEMATICAL MODEL

1. *Navier–Stokes equations in a noninertial coordinate system.* Flow around the rotatable shrouded blade is calculated from the system of Navier–Stokes equations for a compressible gas written in the noninertial rotatable coordinate system in which the blade in a gas flow is fixed, while the shroud (ring) rotates.

We will denote the absolute velocity vector in the original fixed coordinate system by $\mathbf{u}$ and introduce a movable coordinate system rotating at a constant angular velocity $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)^T$ about an axis fixed in the original coordinate system.

We will define the relative velocity vector as follows:

$$\mathbf{u}' = \mathbf{u} - \mathbf{V}, \quad \mathbf{V} = \boldsymbol{\omega} \times \mathbf{r}.$$

We will point out two important properties of the vector of the linear velocity of rotation.

1. The vector $\mathbf{V}$ gradient, that is, the dyadic product of the vector operator ∇ and the vector $\mathbf{V}$ is the skew-symmetric tensor

$$\nabla \mathbf{V} = \nabla \otimes \mathbf{V} = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & \omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}.$$

From this expression it follows that

$$\text{div } \mathbf{V} = 0, \quad \boldsymbol{\omega} = \frac{1}{2} \text{curl } \mathbf{V}, \quad (3.1)$$

while its symmetric part is zero

$$\frac{1}{2} \left(\frac{\partial V_i}{\partial x_j} + \frac{\partial V_j}{\partial x_i} \right) = 0. \quad (3.2)$$

2. It can easily be seen that the vector $\mathbf{V}$ satisfies the following differential relations

$$(\mathbf{V} \cdot \nabla) \mathbf{V} = \boldsymbol{\omega} \times \mathbf{V} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}), \quad (\mathbf{u}' \cdot \nabla) \mathbf{V} = \boldsymbol{\omega} \times \mathbf{u}'. \quad (3.3)$$

The system of Navier–Stokes equations written in the rotatable coordinate system in the form of conservation laws with respect to the velocity $\mathbf{u}'$ is as follows (see, for example, [11]):

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \text{div } \rho \mathbf{u}' &= 0, \\ \frac{\partial \rho \mathbf{u}'}{\partial t} + \text{div } \rho \mathbf{u}' \otimes \mathbf{u}' + \nabla p &= \text{div } \mathbf{S} + \rho \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) - 2\rho (\boldsymbol{\omega} \times \mathbf{u}'), \\ \frac{\partial E}{\partial t} + \text{div}(E + p) \mathbf{u}' &= \text{div } \mathbf{q} + \text{div } \mathbf{S} \mathbf{u}', \end{aligned} \quad (3.4)$$

where $E = \rho \mathbf{u}'^2 + \rho \varepsilon$ is the total energy and ε is the internal energy of the gas. System (3.4) is closed by the equation of state for the perfect gas $p = (\gamma - 1)\rho \varepsilon$. In system (3.4) $\mathbf{S}$ is the strain rate tensor defined as

$$S_{ij} = 2\mu \frac{1}{2} \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right) - \delta_{ij} \frac{2}{3} \mu \frac{\partial u'_i}{\partial x_i} \quad (3.5)$$

and the vector $\mathbf{q}$ is the total energy flux¹

$$q_i = \frac{\mu}{\gamma \text{Pr}} \frac{\partial \varepsilon}{\partial x_i},$$

where the Prandtl number is given by the relation $\text{Pr} = \mu c_p / \lambda$, μ is dynamic viscosity of the gas, c_p is the specific heat at constant pressure, λ is thermal conductivity, and γ is the adiabatic exponent.

Applying Eqs. (3.3) we will bring the system of equations (3.4) into the form using the absolute velocity vector $\mathbf{u}$. We note that, in view of Eqs. (3.1) and (3.2), the strain rate tensor (3.3) is invariant with respect to the rotation velocity of the coordinate system and, therefore, replacing the relative velocity by the absolute velocity does not change the strain rate tensor form given by Eq. (3.5).

Finally, the system of Navier–Stokes equations written in the rotatable coordinate system with respect to the absolute velocity vector reads as follows:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \text{div} \rho(\mathbf{u} - \mathbf{V}) &= 0, \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \text{div} \rho(\mathbf{u} - \mathbf{V}) \otimes \mathbf{u} + \nabla p &= \text{div} \mathbf{S} - \rho(\boldsymbol{\omega} \times \mathbf{u}), \\ \frac{\partial E}{\partial t} + \text{div} \rho(\mathbf{u} - \mathbf{V})E + \text{div} \mathbf{u}p &= \text{div} \mathbf{q} + \text{div} \mathbf{S}\mathbf{u}. \end{aligned} \quad (3.6)$$

From the standpoint of an observer in the fixed coordinate system the system of equations (3.6) describes the variation in the conservative variables at the expense of their translation in the medium rotating at the velocity $\mathbf{V}$, the pressure gradient, and the velocity vector rotation by an angle equal to $|\boldsymbol{\omega}|t$. In numerically realizing this system of equations the rotation velocity may be interpreted as the movable grid velocity. In this form, the system of equations was considered, for example, in [12, 13], where the calculations were simultaneously conducted in both rotatable and fixed domains.

We will rewrite system (3.6) in the pseudovector flux form:

$$\begin{aligned} \frac{\partial \mathbf{Q}}{\partial t} + \frac{\partial \mathbf{F}_x^I}{\partial x} + \frac{\partial \mathbf{F}_y^I}{\partial y} + \frac{\partial \mathbf{F}_z^I}{\partial z} - \frac{\partial \mathbf{F}_x^{NI}}{\partial x} - \frac{\partial \mathbf{F}_y^{NI}}{\partial y} - \frac{\partial \mathbf{F}_z^{NI}}{\partial z} &= \mathbf{H} + \frac{\partial \mathbf{F}_x^V}{\partial x} + \frac{\partial \mathbf{F}_y^V}{\partial y} + \frac{\partial \mathbf{F}_z^V}{\partial z}, \\ \mathbf{Q} &= (\rho, \rho u, \rho v, \rho w, e)^T, \\ \mathbf{H} &= (0, -v\omega_z + w\omega_y, u\omega_z - w\omega_x, -u\omega_y + v\omega_x, 0)^T, \end{aligned} \quad (3.7)$$

where $\mathbf{F}_x^I$, $\mathbf{F}_y^I$, and $\mathbf{F}_z^I$ are the convective fluxes of the system of Navier–Stokes equations in the inertial (fixed) coordinate system, $\mathbf{F}_x^{NI} = V_x \mathbf{Q}$, $\mathbf{F}_y^{NI} = V_y \mathbf{Q}$, and $\mathbf{F}_z^{NI} = V_z \mathbf{Q}$ are the additions to the fluxes determined by the noninertial coordinate system or the grid motion velocity, and $\mathbf{F}_x^V$, $\mathbf{F}_y^V$, and $\mathbf{F}_z^V$ are the viscous fluxes of the system of Navier–Stokes equations determined by the gradients of the absolute velocity vector and the heat flux.

The Jacobi matrices of the convective flux initiated by the rotation velocity have the diagonal form:

$$\partial \mathbf{F}_{x,y,z}^{NI} / \partial \mathbf{Q} = V_{x,y,z} \mathbf{I}.$$

¹We consider a polytropic gas, which means that the specific heat c_V is temperature-independent and, therefore, the internal energy is the linear function of the temperature: $\varepsilon = c_V T$.

Therefore, the Jacobi matrices of the total convective flux can be written in the form:

$$\frac{\partial(\mathbf{F}'_{x,y,z} - \mathbf{F}^{NI}_{x,y,z})}{\partial \mathbf{Q}} = \mathbf{S}_{x,y,z} \text{diag}(\lambda^i_{x,y,z} - V_{x,y,z}) \mathbf{S}_{x,y,z}^{-1}, \quad i = 1, \dots, 5,$$

where $\lambda^i_{x,y,z}$ are the eigenvalues and $\mathbf{S}_{x,y,z}$ ($\mathbf{S}_{x,y,z}^{-1}$) are the matrices of the right (left) eigenvectors of the Jacobi matrices $\partial(\mathbf{F}'_{x,y,z} - \mathbf{F}^{NI}_{x,y,z})/\partial \mathbf{Q}$ of the corresponding convective fluxes of the Navier–Stokes equations in the inertial coordinate system. Thus, the difference of system (3.7) from that of Navier–Stokes equations written in the fixed coordinate system is only in the variation in the characteristic velocities and the presence of the source term on the right side. For this reason, in numerically realizing system (3.7) by means of the Godunov-type schemes the approximation methods remain the same as in solving the equations in the fixed coordinate system.

2. Boundary conditions. For the system of Navier–Stokes equations the boundary no-slip conditions are imposed on solid boundaries. These consist in equating the relative velocities (in the rotating noninertial coordinate system) at the boundaries to zero. In the case of employing the system of equations (3.6) written with respect to the absolute velocity vector (in the fixed coordinate system) the no-slip conditions are written as follows:

$$\mathbf{u} - \mathbf{V} = 0$$

for the rotating blade, where $\mathbf{V}$ is the linear velocity of the surface rotation, and

$$\mathbf{u} = 0$$

for the fixed shroud (ring).

Since the numerical simulation of the problem is performed in a bounded region, some artificial boundary conditions must be imposed on the computation domain boundary.

These boundary conditions are determined by splitting the fluxes relating the values of the gasdynamic parameters ρ_i , $\mathbf{U}_i$, and p_i within the computation domain and their values in the distant flow ρ_∞ , $\mathbf{U}_\infty$, and p_∞ in the directions of the characteristic velocities. The latter values are determined by the characteristic relations for an isentropic gas

$$\rho_\infty = \rho_i \left(\frac{p_\infty}{p_i} \right)^{1/\gamma}, \quad p_\infty = p_0, \quad \mathbf{U}_\infty = \mathbf{U}_i + \frac{2}{\gamma - 1} \left[\left(\gamma \frac{p_i}{\rho_i} \right)^{1/2} - \left(\gamma \frac{p_\infty}{\rho_\infty} \right)^{1/2} \right] \cdot \mathbf{n},$$

where p_0 is the constant pressure of the undisturbed gas and $\mathbf{n}$ is the vector of the unit outward normal to the computation domain boundary.

3. Modeling the far field acoustics. Modeling the acoustics in the far flowfield is considered to mean the technique of calculating the pressure fluctuations at an arbitrarily distant point. The mathematical model describing the acoustic disturbance propagation in the far field, that is, in an infinite domain encompassing the near field that contains the acoustic source, is given by the linearized Euler equations which degenerate into the wave equation in the case of an isentropic flow.

In modeling the far-field pressure fluctuations the boundary conditions are specified on a certain control surface. The necessary values on these surface are preassigned from the solution of the problem for the near flowfield. The latter is calculated in an extended domain including the part of space encompassed by the control surface. Thus, the control surface is embedded in the computation grid of the near-field computation domain. As a result, the far field problem is governed by the linearized equations in the domain exterior to the control surface, while the boundary conditions for the problem are preassigned on the control surface.

Generally, the solution of this problem is presented by the Ffowcs Williams–Hawkings (FWH) formula [7] which generalizes the integral Kirchhof formula to the case of a nonuniform background field and a time-dependent control surface.

We will describe the technique of selecting the control surface. We assume that any point of the control surface S_0 with the outward unit normal $\mathbf{n}$ is parametrized by a vector $\mathbf{x}_0$ and its motion in the coordinate

system fitted to the background flow is given by the dependence $\mathbf{f}(\mathbf{x}_0, t)$ (source coordinate). We seek the solution of the problem at point $\mathbf{x}$ (observer's coordinate) at the moment of time t . We introduce the dimensionless velocity of the background flow relative to the point on the control surface

$$\mathbf{M} = -\frac{1}{c_0} \frac{\partial \mathbf{f}}{\partial t}(\mathbf{x}_0, t) \quad (3.8)$$

and determine the radiation vector from the equation $\mathbf{r}_e = \mathbf{x} - \mathbf{f}(\mathbf{x}_0, t) + \mathbf{M}\boldsymbol{\tau}_e$, where $\boldsymbol{\tau}_e = -|\mathbf{r}|_e/c_0$ is the delay time. We will write down the FWH formula neglecting the terms which decrease as r^{-2} with increase in the radius r

$$p'(\mathbf{x}, t) = \frac{1}{4\pi} \int_{S_0} \left[\frac{1}{(1 - M_r)^2 r} \frac{\partial}{\partial r} (c_0 \rho' (M_n - n_r) + \rho_0 (u_n - u_r M_n)) \right]_{\tau=t+\tau_e, \mathbf{r}=\mathbf{r}_e} ds, \quad (3.9)$$

where the velocity vector $\mathbf{u}$ is taken with respect to the background flow, while the subscripts n and r refer to the projections of the corresponding vectors on the normal and the radiation vector direction.

4. NUMERICAL METHOD

1. *Numerical method for calculating the near flowfield.* The method of calculating the near flowfield is based on the solution of the system of Navier–Stokes equations in the noninertial coordinate system (3.7). The spatial approximation of the convective fluxes of this system on tetrahedral grids is realized using a scheme based on the quasi-one-dimensional reconstruction of the variables along a grid edge (EBR scheme). This class of schemes is described in detail in [7–9]. We note that the spatial discretization is based on the vertex-centered formulation, which means that all the unknown variables are determined at gridpoints surrounded by computation cells (dual grid).

On the tetrahedral grids, for which the dual grid has computation cells of the same shape (analogue of the uniform Cartesian grid), the scheme based on the quasi-one-dimensional reconstruction along a grid edge can be of as high as fifth and even sixth order in space at the corresponding reconstruction of the flux variables [8, 9]. In the case of an arbitrary unstructured tetrahedral grid and the same reconstruction the approximation order is formally only second. However, this scheme ensures a higher accuracy (from the standpoint of the difference between the numerical and exact solutions with respect to the grid norm) than the other second-order schemes [7].

The spatial approximation of the viscous terms in the system of Navier–Stokes equations is realized using the finite-element Galerkin method with linear basis functions.

The time integration is performed using an implicit three-layer second-order scheme followed by the Newtonian linearization of the finite-difference (in space) system of equations. At each Newtonian iteration stage the method of biconjugate gradients (BiCGSTAB) [14] is applied for solving the system of linear equations.

The method is realized in the NOISEtte software package [15].

2. *Calculation of acoustic disturbances in the far field.* To calculate pressure fluctuations at an arbitrary distant point by the FWH formula (3.9) it is necessary to preassign a control surface on which the flowfield is determined from the calculations of the system of Navier–Stokes equations. We will consider different approaches to the selection of the control surface. Frequently, a control surface coinciding with the blade is used. However, this approach does not account for the effect of flow nonuniformity near the blade on the generated sound propagation and in the case of a shrouded rotor it misjudges the ring effect on the noise propagation.

In many publications concerned with the far field calculations (see, for example, [16]) the control surface is rigidly fitted to the grid surrounding the blade, that is, the points on the surface rotate together with the blade. However, in this case the rotation velocity of the points on the control surface calculated from Eq. (3.8) turns out to be ωr_s , where r_s is the control surface radius. Since r_s can be considerably greater

than the blade radius, the velocity can be similar in value with the speed of sound and even greater than it. This would lead to considerable difficulties in view of a singularity in Eq. (3.9).

To avoid the above-mentioned problems the control surface is fixed in the fixed coordinate system. Thus, a very simple case ($\mathbf{f}(\mathbf{x}_0, t) = \mathbf{x}_0$) is used, where the integral Kirchhof formulas and FWH take the form:

$$p'(\mathbf{x}, t) = \frac{1}{4\pi} \int_{S_0} \left[\frac{1}{r} \frac{\partial}{\partial r} (\rho_0 u_n - c_0 \rho' n_r) \right]_{\tau=t+|\mathbf{x}-\mathbf{x}_0|/c_0, \mathbf{r}=\mathbf{x}-\mathbf{x}_0} ds. \quad (4.1)$$

For the simplicity of the following operations we will take a surface of revolution parametrized by the angles θ and ψ as the control surface. In this case, we have

$$\mathbf{x}_0(\theta, \psi) = \begin{pmatrix} r(\theta) \cos \theta \cos \psi \\ r(\theta) \cos \theta \sin \psi \\ r(\theta) \sin \theta \end{pmatrix}. \quad (4.2)$$

The choice of the function $r(\theta)$ determines the control surface shape.

If the calculations of the near field indicate that in the blade-fitted coordinate system the physical fields are stationary, then formula (4.1) can be simplified. In the background flow system the pressure and the radial velocity depend only on the rotation of the auxiliary surface relative to the control surface

$$\rho(t, \theta, \psi) = \rho(0, \theta, \psi - \omega t), \quad u_n(t, \theta, \psi) = u_n(0, \theta, \psi - \omega t). \quad (4.3)$$

For this reason, the time derivatives of the pressure and the radial velocity can be expressed in terms of the derivatives with respect to the angle ψ , as follows:

$$\frac{\partial \rho}{\partial t}(t, \theta, \psi) = -\omega \frac{\partial \rho}{\partial \psi}(0, \theta, \psi - \omega t), \quad \frac{\partial u_n}{\partial t}(t, \theta, \psi) = -\omega \frac{\partial u_n}{\partial \psi}(0, \theta, \psi - \omega t).$$

Thus, the formula for calculating the pressure fluctuations in the far field takes the form:

$$p'(\mathbf{x}, t) = \frac{\omega}{4\pi|\mathbf{x}|} \int_{S_0} \left(\rho_0 \frac{\partial u_n}{\partial \psi} - c_0 \frac{\partial p'}{\partial \psi} \frac{r_n}{r} \right) \left(t, \theta, \psi + \omega \left(t - \frac{r}{c_0} \right) \right) ds. \quad (4.4)$$

The derivatives with respect to ψ are calculated as the derivatives from a quadratic interpolant on a uniform surface grid.

For any point $\mathbf{x}$ in the far field integral (4.4) was calculated at 240 points in time with the step of $2\pi/(120\omega)$ thus covering the signal period. The calculation of the signal at additional points does not provide an additional information on it.

On the basis of the signal calculated at a far-field point its spectrum is determined on the basis of the discrete Fourier transformation.

3. *Computation grid.* As mentioned in Sect. 1, the computation domain of the problem is a sector containing one blade. The outer boundary of the computation domain is so constructed that it describes the flow region produced by the rotor both directly beneath it (along the axis of rotation) and where it is formed above the rotor. The computation domain dimensions outside the ring are so chosen that it is possible to analyze the flow parameters outside the ring and to construct a control surface whose size would be sufficient to analyze the acoustic properties of the shrouded rotor (Fig. 4a).

An unstructured tetrahedral grid is constructed in the computation domain. It is refined in the regions, where the detailed modeling of the aerodynamic effects is required. Thus, the most detailed grid is constructed on the leading and trailing (along the airfoil chord) regions of the blade surface and on the blade endface. On the surfaces of the blade, the central body, and the ring the grid is refined in the normal-to-surface directions in order to resolve the boundary layer.

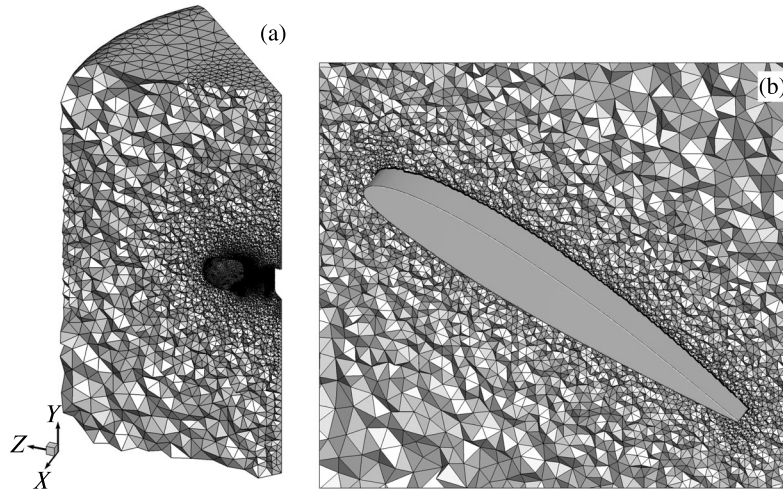


Fig. 4. General view of the computation grid; longitudinal section (a) and grid near the blade (b).

Computation grid dimensions for different pitch angles

Pitch angle α , deg	Gridpoint number	Number of tetrahedrons
10	2326941	13589791
15	2371707	13859009
20	2511582	14669900
30	2555652	14936402
40	2592469	15154418

Then the surface and volume grids are constructed using the advancing front technique, so that the grid element dimensions vary smoothly (Fig. 4). The most detailed grid is constructed in the gap region between the blade endface and the inside surface of the ring. Constraints from above on maximum dimensions of the elements near the blade and the central body are also introduced.

On the solid surfaces of the rotor, the central body, and the ring the grid is refined in order to resolve the boundary layer, in accordance with the Reynolds number so that not less than ten cells fall on the boundary layer thickness.

As a result, five computation grids for the blade pitch angles of 10, 15, 20, 30, and 40° were constructed; their dimensions are given in table.

The computation grids were constructed using the ICEM CFD program from the ANSYS software package [17].

5. RESULTS OF THE CALCULATIONS

1. *General flow pattern.* In the calculations a stationary, that is, independent of the angle of rotation ψ , solution was obtained.

In Fig. 5 the flow pattern is presented in the fixed coordinate system for the configuration with the blade pitch angle of 30°. In plotting this figure the complete flow pattern was obtained by continuing the solution obtained for a single sector with the appropriate rotation of the coordinates of each sector and the velocity vector in the sector by an angle $2\pi k/4$, where k varied from 0 to 3.

As can be seen in Fig. 5, the streamlines in the flow ahead of the rotor are directed along the axis inside the ring and are gradually curving toward the ring, so that the effect of air suction from the ambient at rest in the channel formed by the ring is observable. The suction velocity diminishes along the ring radius and becomes negligibly small on the outside surface of the shroud. In a flow behind the rotor the axial flow velocity considerably increases on a cylindrical surface of the order of the ring radius and the flow is swirled in the direction of the rotor rotation. With decrease in the radius and increase in the distance the

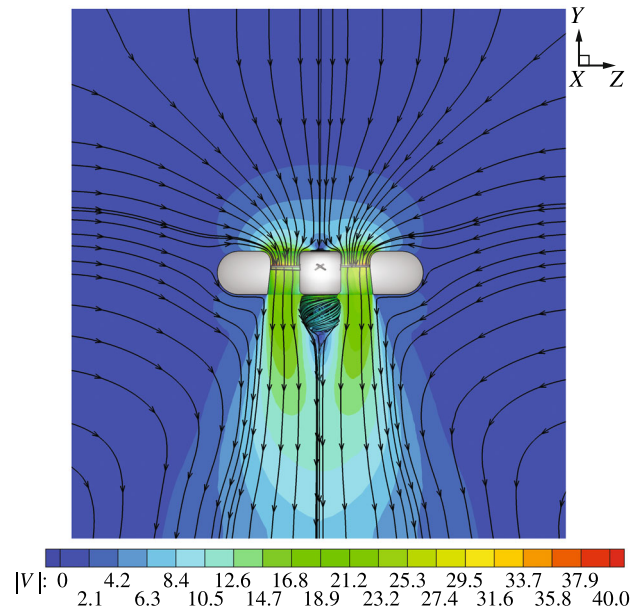


Fig. 5. General view of the flow; streamlines in the fixed coordinate system and the field of the absolute value of the velocity in the meridional section for the blade pitch angle of 30° .

axial velocity diminishes and, as a result, the flow swirling increases. The streamline roll-up into a cord is particularly clearly visible beneath the central body, where the axial velocity is almost zero. This means the formation of helical vortices in the lower flow region, in the vicinity of the axis of rotation.

Figure 6 presents the streamlines and the fields of the absolute value of the absolute velocity in the meridional section along the rotor blade axis for different blade pitch angles. Clearly that the greatest flow velocity can be observable in the jet directly beneath the rotor disk, the flow velocity rapidly decreasing with the distance from the ring. An increase in the velocity and the jet width behind the rotor with increase in the blade pitch angle is also observable, although the general flow pattern does not experience considerable variations: a suction zone is formed in the upper half-plane above the ring, a jet outflow zone occurs in the lower half-plane, directly beneath the rotor disk, and the zone of a cocurrent flow induced by the jet flow exists far away from the axis of rotation.

2. Aerodynamic forces. In Fig. 7 we have plotted the thrust (Fig. 7a) and torque (Fig. 7b) coefficients obtained in the calculations and in the experiment against the blade pitch angle. Obviously, with increase in the pitch angle the axial velocity of the expelled air body behind the rotor increases, while the pressure in the region above the blade reduces. Accordingly, the thrust and the torque coefficient increase. It can be seen that the calculated coefficients are similar in value with the measured ones for all pitch angles for which the calculations were carried out. Good agreement with the experiment is also observable in Fig. 8 in which the rotor polar (thrust coefficient–torque coefficient relation) is plotted. Clearly that the calculated data well fall on the experimental curve.

3. Acoustic fluctuations in the far field. The acoustic parameters of the shrouded rotor were analyzed using the integral formula of the FWH type (4.4). The adequacy of this approach within the framework of the NOISEtte software package was earlier confirmed by means of comparing with the exact solution of the problem of the point source acoustics in subsonic flow [8]. In [8] the data on the validation of the technique are presented with reference to the example of the problem of sound generation in subsonic flow past tandem cylinders.

In the problem under consideration three versions of the control surface given by the parametric function (4.2) were chosen, namely, surface 1 encompassing the region directly around the shrouded rotor and the zone of flow blown out of the ring, surface 2 including the region of flow sucked by the ring, and surface 3 encompassing the zones both above and beneath the ring (Fig. 9a).

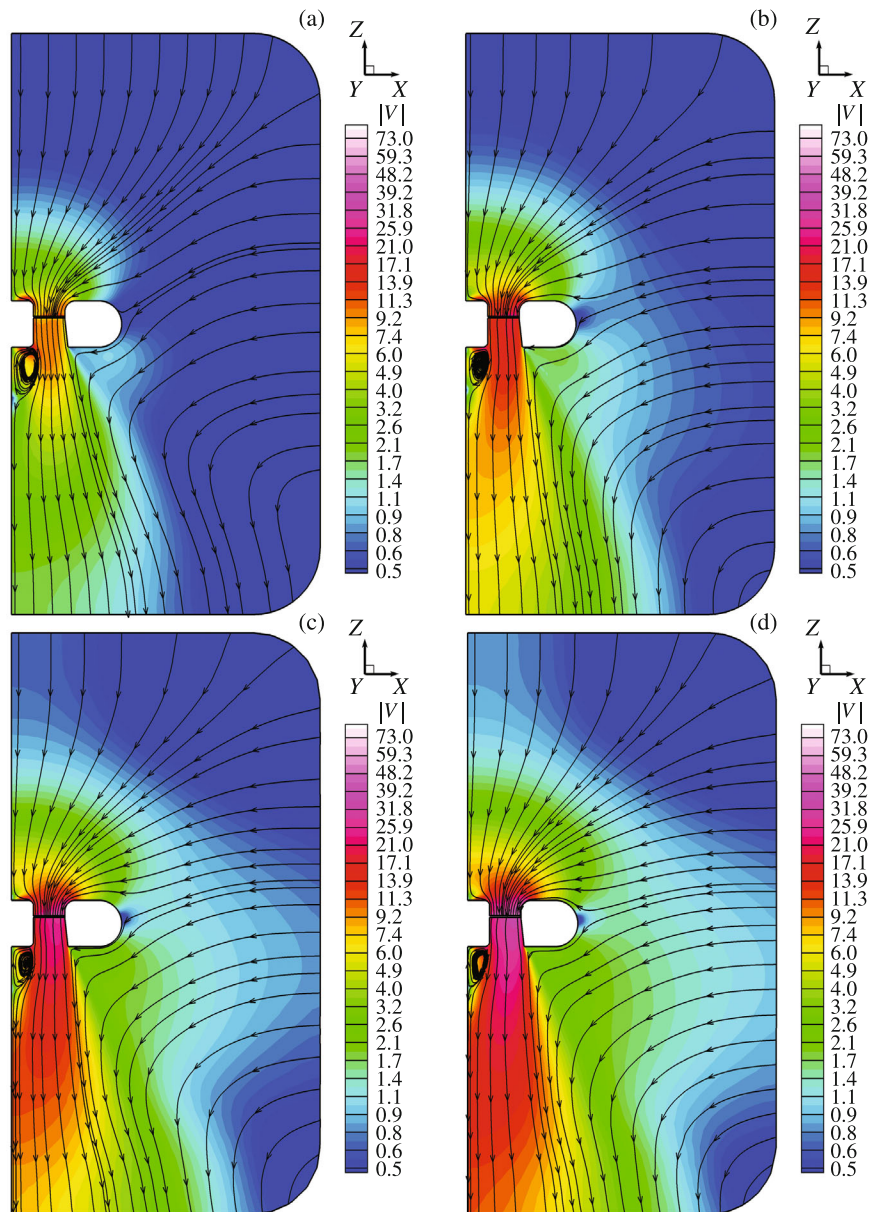


Fig. 6. Streamlines and the field of the absolute value of the absolute velocity in the meridional section for the blade pitch angles of 10° (a), 20° (b), 30° (c), and 40° (d).

As follows from Eq. (4.3), in the stationary regime, that is, when the flowfield in the rotor-fitted coordinate system is time-independent, time T appears only in the angle of rotation. This is physically meaningful, since the signal variation with time is due only to the rotation of all the flowfields around the rotor axis. Since the period of the angle is 2π , the rotor-radiated acoustic signal is also periodic, with the period $2\pi/\omega$. Moreover, since all the rotor blades are the same, the flow period is $2\pi/N$, where N is the blade number, while the signal period is $2\pi/(N\omega)$.

The periodic signal spectrum is discrete, the minimum frequency being inverse proportional to the signal period and equal to $N\omega$, and the other frequencies being multiple to the minimum frequency. If the flowfield is fairly smooth, it might be expected that it is the lower modes that possess the greatest intensity and their amplitudes decrease with increase in the mode number. This is confirmed by the numerical experiment.

In Fig. 10 we have plotted the radiation patterns for the overall sound pressure level at a distance $R = 150$ m in the polar coordinate system in the radiation pattern plane for different control surfaces. The ra-

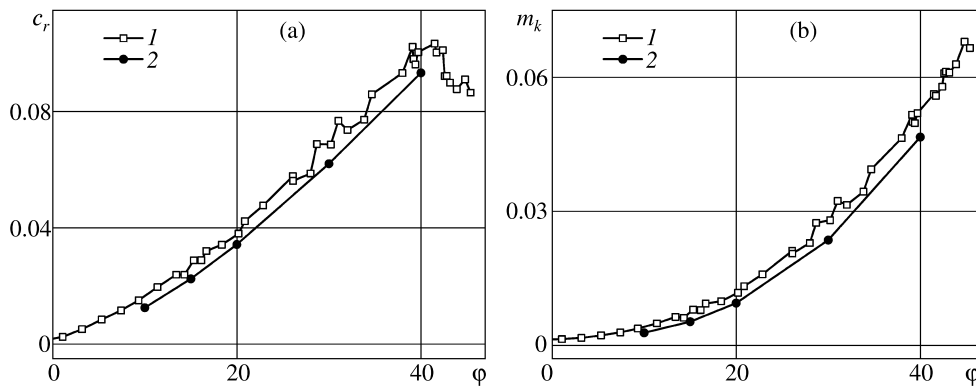


Fig. 7. Rotor torque (a) and thrust (b) coefficients; (1) experiment and (2) calculations.

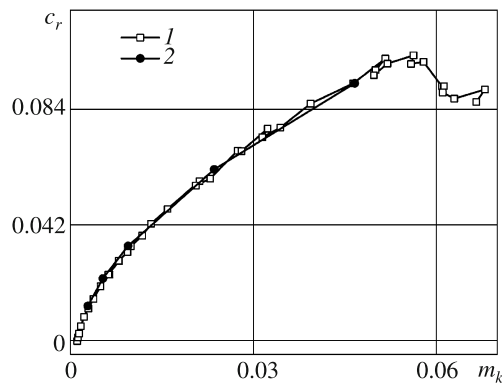


Fig. 8. Rotor polar; (1) experiment and (2) calculations.

diation pattern plane is perpendicular to the rotor rotation plane, the choice of the plane being independent of the polar angle ψ due to the symmetry. The angular coordinate is given by the azimuthal angle $\beta = \theta + \pi/2$ (Fig. 9b). The spectral power of the pressure fluctuations is measured along the radial axis.

The general analysis of the radiation patterns presented in Fig. 10 shows that the greatest level of the noise produced by the rotating shrouded rotor is observable behind the ring in the plane of rotor rotation and in its vicinity. The weakest sound is detected at the axis of rotor rotation. This indicates that this design possesses the least possible acoustics signature for observers directly beneath or above the rotor. Contrariwise, for observers on the side of the shrouded rotor the noise intensity is maximum.

It can also be seen that the larger control surface (surface 3) makes it possible to more adequately describe the rotor-generated acoustic field, compared with the results obtained using surfaces 1 and 2, the peak of the acoustic radiation power with respect to the azimuthal angle falls on the rotor-produced jet region ($\beta \approx 75^\circ$) and is considerably higher, when the control surface 3 is employed (Fig. 10).

We note that with increase in the blade pitch angle the overall power of the sound radiation increases, while its peak is displaced into the jet flow region behind the rotor. It is also noteworthy that, according to the calculated results, the overall power of the sound radiation increases with the rotor rotation velocity but in this case a noticeable displacement of the power peak into the jet region is not observable (Fig. 10).

An important acoustic parameter is also the noise decay rate, as the observer recedes. Basing on the adopted model of the far field acoustics, Eq. (4.4) takes account only for the contribution of the terms most slowly decreasing with the distance, that is, the terms corresponding to the sound decay rate inverse proportional to the distance to the observer. Because of this, given the radiation pattern at the control distance of 150 m, one can easily estimate the sound pressure at the most distant points on the field. The calculations on the above-mentioned grids were carried out using the computational structures of the Joint Supercomputer Center of the Russian Academy of Sciences (MVS-10P cluster [18]). The calculations were performed using the NOISEtte software package in the parallel regime using 200 to 480 computational

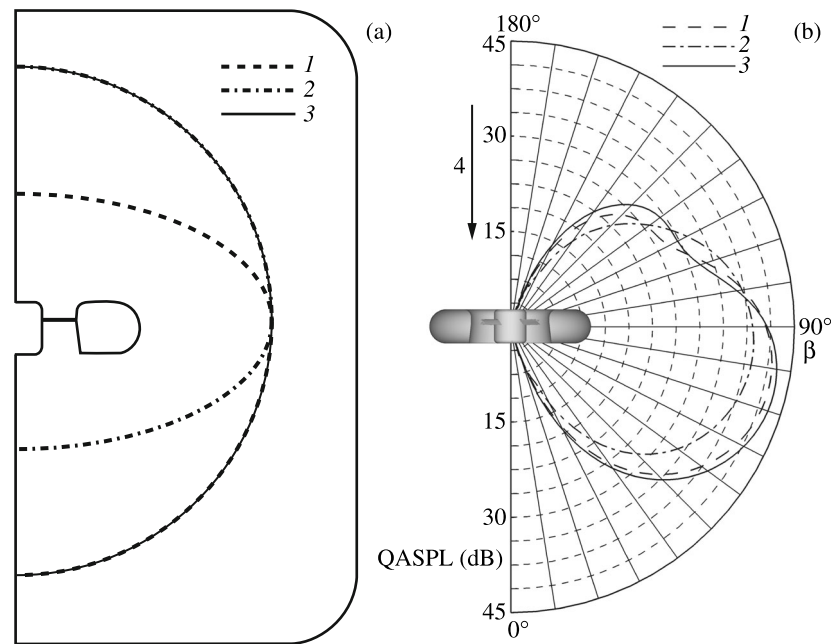


Fig. 9. Control surfaces for the FWH method (a) and the plane of the radiation pattern for the blade pitch angle of 40° ($0 \leq \beta \leq \pi$) (b); (1, 2, and 3) are the control surfaces and (4) is the rotor-induced jet direction.

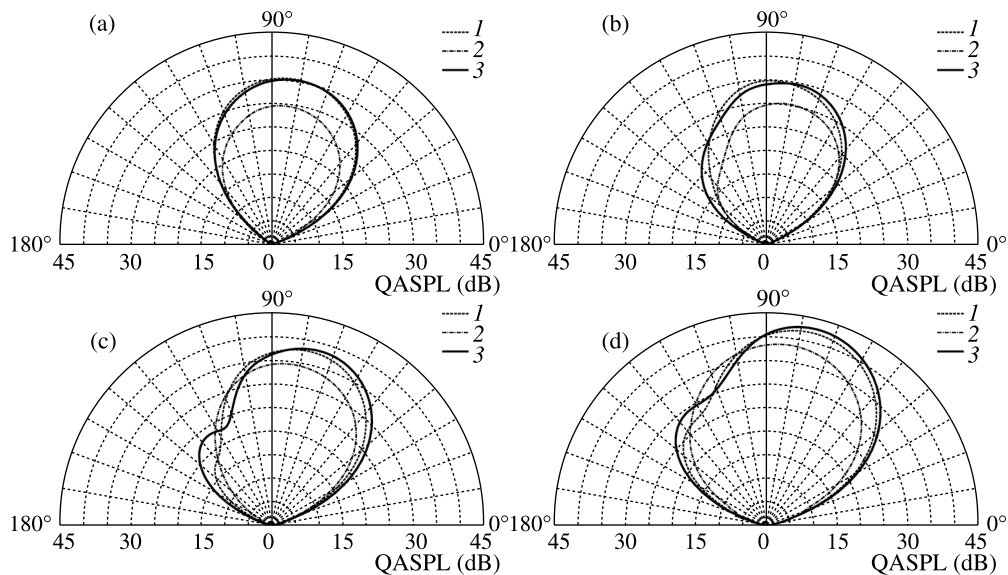


Fig. 10. Radiation patterns for different blade pitch angles and different control surfaces (1, 2, and 3); 10° (a), 15° (b), 30° (c), and 40° (d).

cores with the two-level MPI+OpenMP parallelization [15]. The calculations of one configuration for a fixed blade pitch angle at a fixed rotor rotation velocity took about one day of the computer time.

Summary. Viscous gas flow generated around a model shrouded tail rotor of a helicopter is numerically investigated. The numerical technique developed is validated against the numerical and experimental data on the aerodynamic characteristics of the rotor in the absence of outer flow (hovering operation regime). On the basis of the numerical results on the near field flow representing a distributed acoustic source the parameters of the far-field acoustic radiation of the configuration considered are calculated. In particular, the radiation pattern and its dependence on the blade pitch angle are analyzed.

It should be noted that the computation grids used in this study make it possible to obtain only stationary flow and, accordingly, to use a simplified model of the far field acoustics which accounts for multiple discrete

tones of the acoustic field produced by the rotor rotation. At the same time, the broadband eddy noise caused by vortical interactions and small-scale turbulence remains beyond the scope of the study. Later, by means of conducting selective calculations on detailed grids we are planning to investigate the broadband component effect on the acoustic radiation power and its spectral composition in the far field. Supposedly, as applied to the shrouded rotor configuration considered the broadband noise is not particularly noticeable in the far field. However, its adequate modeling is of key importance in numerically investigating the main helicopter rotor and, possibly, actual tail rotor configurations.

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